1.56 The Binomial Distribution

Worked Solutions

1)

$X \sim \mathrm{B}(6, 0.2)$ is diagram B

$X \sim \mathrm{B}(6, 0.5)$ is diagram C

$X \sim \mathrm{B}(6, 0.7)$ is diagram A

$X \sim \mathrm{B}(6, 0.89)$ is diagram D

2 a)

Positive skew

2 b)

Symmetric

2 c)

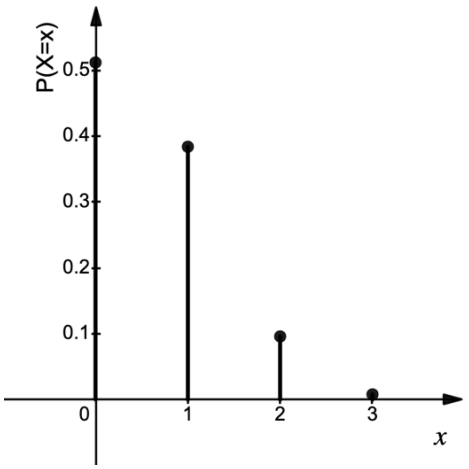
Negative skew

2 d)

Negative skew

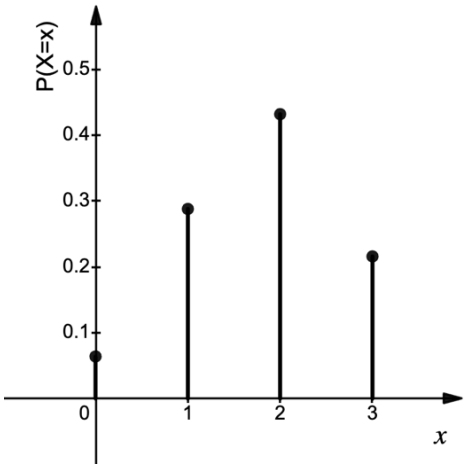
3)

x	0	1	2	3
$\mathrm{P}(X = x)$	0.512	0.384	0.096	0.008



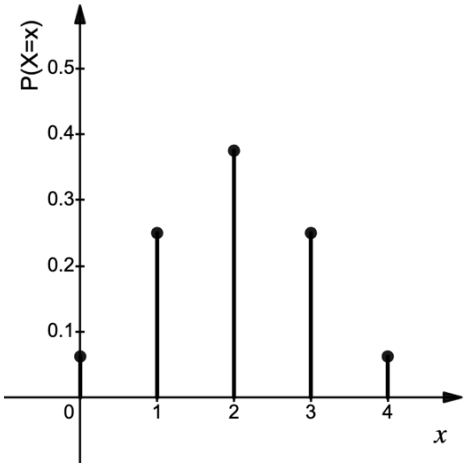
4)

x	0	1	2	3
$\mathrm{P}(X = x)$	0.064	0.288	0.432	0.216



5)

x	0	1	2	3	4
$\mathrm{P}(X = x)$	0.0625	0.25	0.375	0.25	0.0625



6)

$X \sim \mathrm{B}(10, 0.5)$

7 a)

$\binom{15}{3} \times 0.3^3 \times 0.7^{12} = 0.1700$ (4 DP)

7 b)

$\binom{15}{4} \times 0.3^4 \times 0.7^{11} = 0.2186$ (4 DP)

7 c)

$\binom{15}{5} \times 0.3^5 \times 0.7^{10} = 0.2061$ (4 DP)

7 d)

$\binom{15}{9} \times 0.3^9 \times 0.7^6 = 0.0116$ (4 DP)

8 a)

$\binom{10}{6} \times 0.6^6 \times 0.4^4 = 0.2508$ (4 DP)

8 b)

$\binom{25}{4} \times 0.2^4 \times 0.8^{21} = 0.1867$ (4 DP)

8 c)

$\binom{60}{57} \times 0.9^{57} \times 0.1^3 = 0.0844$ (4 DP)

8 d)

$\binom{105}{52} \times 0.56^{52} \times 0.44^{53} = 0.0320$ (4 DP)

9 a)

0.1029

9 b)

0.2668

9 c)

0.0282

10 a)

0.0576

10 b)

0.2054

10 c)

0.1746

11 a)

$\mathrm{P}(X = 0) = \binom{17}{0} \times 0.16^0 \times 0.84^{17} = 0.05161166\ldots$

$\mathrm{P}(X = 1) = \binom{17}{1} \times 0.16^1 \times 0.84^{16} = 0.16712348\ldots$

$\mathrm{P}(X \leq 1) = \mathrm{P}(X = 0) + \mathrm{P}(X = 1)$

$= 0.0516\ldots + 0.1671\ldots = 0.2187$ (4 DP)

11 b)

$\mathrm{P}(X = 2) = \binom{17}{2} \times 0.16^2 \times 0.84^{15} = 0.25466435\ldots$

$\mathrm{P}(X \leq 2) = \mathrm{P}(X = 0) + \mathrm{P}(X = 1) + \mathrm{P}(X = 2)$

$= 0.0516\ldots + 0.1671\ldots + 0.2546\ldots = 0.4734$ (4 DP)

11 c)

$\mathrm{P}(X = 3) = \binom{17}{3} \times 0.16^3 \times 0.84^{14} = 0.24253748\ldots$

$\mathrm{P}(X \leq 2) = \mathrm{P}(X = 0) + \mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \mathrm{P}(X = 3)$

$= 0.0516\ldots + 0.1671\ldots + 0.2546\ldots + 0.2425\ldots$

$= 0.7159$ (4 DP)

12 a)

0.0061

12 b)

0.1629

12 c)

0.6466

13 a)

0.7512

13 b)

0.9884

13 c)

0.6720

13 d)

0.1713

14 a)

$\mathrm{P}(X < 9) = \mathrm{P}(X \leq 8) = 0.0329$

14 b)

$\mathrm{P}(X < 11) = \mathrm{P}(X \leq 10) = 0.1878$

14 c)

$\mathrm{P}(X < 14) = \mathrm{P}(X \leq 13) = 0.7309$

15 a)

$\mathrm{P}(X \geq 8) = 1 - \mathrm{P}(X \leq 7) = 0.6552$

15 b)

$\mathrm{P}(X \geq 11) = 1 - \mathrm{P}(X \leq 10) = 0.1462$

15 c)

$\mathrm{P}(X \geq 12) = 1 - \mathrm{P}(X \leq 11) = 0.0611$

16 a)

$\mathrm{P}(X \geq 4) = 1 - \mathrm{P}(X \leq 3) = 0.8982$

16 b)

$\mathrm{P}(X \geq 2) = 1 - \mathrm{P}(X \leq 1) = 0.1760$

16 c)

$\mathrm{P}(X \geq 52) = 1 - \mathrm{P}(X \leq 51) = 0.8158$

17 a)

$\mathrm{P}(9 \leq X \leq 12) = \mathrm{P}(X \leq 12) - \mathrm{P}(X \leq 8)$

$= 0.120338\ldots - 0.002190\ldots = 0.1181$

17 b)

$\mathrm{P}(10 \leq X \leq 14) = \mathrm{P}(X \leq 14) - \mathrm{P}(X \leq 9)$

$= 0.374483\ldots - 0.007547\ldots = 0.3669$

17 c)

$\mathrm{P}(12 \leq X \leq 14) = \mathrm{P}(X \leq 14) - \mathrm{P}(X \leq 11)$

$= 0.374483\ldots - 0.055505\ldots = 0.3190$

18 a)

$\mathrm{P}(2 < X \leq 3) = \mathrm{P}(X \leq 3) - \mathrm{P}(X \leq 2)$

$= 0.340883\ldots - 0.157006\ldots = 0.1839$

18 b)

$\mathrm{P}(3 < X \leq 7) = \mathrm{P}(X \leq 7) - \mathrm{P}(X \leq 3)$

$= 0.954219\ldots - 0.340883\ldots = 0.6133$

18 c)

$\mathrm{P}(4 < X \leq 10) = \mathrm{P}(X \leq 10) - \mathrm{P}(X \leq 4)$

$= 0.999138\ldots - 0.558632\ldots = 0.4405$

19 a)

$\mathrm{P}(14 \leq X < 17) = \mathrm{P}(X \leq 16) - \mathrm{P}(X \leq 13)$

$= 0.802398\ldots - 0.396727\ldots = 0.4057$

19 b)

$\mathrm{P}(13 \leq X < 18) = \mathrm{P}(X \leq 17) - \mathrm{P}(X \leq 12)$

$= 0.889356\ldots - 0.263280\ldots = 0.6261$

19 c)

$\mathrm{P}(10 \leq X < 13) = \mathrm{P}(X \leq 12) - \mathrm{P}(X \leq 9)$

$= 0.263280\ldots - 0.038958\ldots = 0.2243$

20 a)

$\mathrm{P}(4 < X < 6) = \mathrm{P}(X \leq 5) - \mathrm{P}(X \leq 4)$

$= 0.426806\ldots - 0.227948\ldots = 0.1989$

20 b)

$\mathrm{P}(5 < X < 10) = \mathrm{P}(X \leq 9) - \mathrm{P}(X \leq 5)$

$= 0.979658\ldots - 0.426806\ldots = 0.5529$

20 c)

$\mathrm{P}(6 < X < 9) = \mathrm{P}(X \leq 8) - \mathrm{P}(X \leq 6)$

$= 0.930150\ldots - 0.643741\ldots = 0.2864$

21)

$X \sim \mathrm{B}(30, 0.2)$

$\mathrm{P}(X = 5) = 0.1723$

22)

$X \sim \mathrm{B}(200, 0.008)$

$\mathrm{P}(X < 3) = \mathrm{P}(X \leq 2) = 0.7838$

23 a)

$\mathrm{E}(X) = np = 4$

$\mathrm{Var}(X) = np(1-p) = 2$

23 b)

$\mathrm{E}(X) = np = 6$

$\mathrm{Var}(X) = np(1-p) = 4.2$

23 c)

$\mathrm{E}(X) = np = 14.4$

$\mathrm{Var}(X) = np(1-p) = 1.44$

23 d)

$\mathrm{E}(X) = np = 5$

$\mathrm{Var}(X) = np(1-p) = 4$

23 e)

$\mathrm{E}(X) = np = 24$

$\mathrm{Var}(X) = np(1-p) = 9.6$

24 a)

$\mathrm{E}(X) = np = 13.2$

$\mathrm{P}(X \leq 13.2) = \mathrm{P}(X \leq 13) = 0.5460$

24 b)

$\mathrm{P}(X > 13.2) = \mathrm{P}(X \geq 14) = 1 - \mathrm{P}(X \leq 13)$

$= 1 - 0.545954\ldots = 0.4540$

25)

$n = 45, p = 0.25$

$\mathrm{E}(X) = np = 11.25$

$\mathrm{Var}(X) = np(1-p) = 8.4375$

26)

$100 = 450p \Rightarrow p = \frac{2}{9}$

27)

$120 = 1560p \Rightarrow p = \frac{1}{13}$

28)

$42 = n \times 0.3 \times 0.7 \Rightarrow n = 200$

29)

$375 = n \times \frac{1}{6} \times \frac{5}{6} \Rightarrow n = 2700$

30)

$35 = \sqrt{n \times \frac{1}{8} \times \frac{7}{8}} \Rightarrow \frac{7}{64}n = 1225 \Rightarrow n = 11200$

31)

$105 = \sqrt{n \times \frac{5}{8} \times \frac{3}{8}} \Rightarrow \frac{15}{64}n = 11025 \Rightarrow n = 47040$

32)

$np = 25.2$ and $np(1-p) = 2.52$

$\therefore 25.2(1-p) = 2.52 \Rightarrow p = 0.9$

$n \times 0.9 = 25.2 \Rightarrow n = 28$

33)

$np = 14.4$ and $np(1-p) = 8.64$

$\therefore 14.4(1-p) = 8.64 \Rightarrow p = 0.4$

$n \times 0.4 = 14.4 \Rightarrow n = 36$

34)

$np = 5$ and $np(1-p) = \frac{60}{13}$

$\therefore 5(1-p) = \frac{60}{13} \Rightarrow p = \frac{1}{13}$

$n \times \frac{1}{13} = 5 \Rightarrow n = 65$

35)

$\mathrm{Var}(X) = 100 \times 0.2 \times 0.8 = 16$

$\mathrm{E}(Y) = 20k$

$16 = 20k \Rightarrow k = 0.8$

36)

$\mathrm{E}(X) = 3\mathrm{E}(Y) \Rightarrow np = 3kp \Rightarrow n = 3k$

$\mathrm{Var}(X) = 2\mathrm{E}(Y) \Rightarrow np(1-p) = 2kp$

$\therefore 3kp(1-p) = 2kp \Rightarrow 3(1-p) = 2 \Rightarrow p = \frac{1}{3}$

37)

$\mathrm{E}(Y) = 3\mathrm{E}(X) \Rightarrow kp = 3np \Rightarrow n = \frac{1}{3}k$

$\mathrm{E}(X) = 5\,\mathrm{Var}(Y) \Rightarrow np = 5kp(1-p)$

$\therefore \frac{1}{3}kp = 5kp(1-p) \Rightarrow \frac{1}{3} = 5(1-p) \Rightarrow p = \frac{14}{15}$

38)

There are fixed number of independent trials.

There are exactly two outcomes of each trial, with the probability of success (or failure) being constant.

39)

There is no fixed number of trials, n.

40)

The sixteen days are consecutive, so this is not a random sample of days throughout the year. If it rains on one day, then this will affect the likelihood of it raining on the next, so the trials are not independent.

41 a)

$X \sim \mathrm{B}(21, 0.82)$

$\mathrm{P}(X \geq 18) = 1 - \mathrm{P}(X \leq 17) = 0.4616$

41 b)

$\mathrm{P}(X < 16) = \mathrm{P}(X \leq 15) = 0.1625$

42 a)

$X \sim \mathrm{B}(35, 0.125)$

$\mathrm{Var}(X) = 35 \times 0.125 \times 0.875 = 3.828125$

42 b)

$\mathrm{P}(X \leq 6) = 0.8605$

42 c)

$\mathrm{P}(4 < X < 8) = \mathrm{P}(X \leq 7) - \mathrm{P}(X \leq 4)$

$= 0.936749\ldots - 0.551277\ldots = 0.3855$

43 a)

$X \sim \mathrm{B}(50, 0.1)$

$\mathrm{E}(X) = 50 \times 0.1 = 5$

43 b)

$\mathrm{Var}(X) = 50 \times 0.1 \times 0.9 = 4.5$

43 c)

$\mathrm{P}(X = 5) = 0.1849$

43 d)

$\mathrm{P}(X \leq 6) = 0.7702$

43 e)

$0.7702^7 = 0.1608$

44 a)

$X \sim \mathrm{B}(12, 0.15)$

$\mathrm{P}(X > 2) = \mathrm{P}(X \geq 3) = 1 - \mathrm{P}(X \leq 2) = 0.2642$

44 b)

$3 \times 0.264181\ldots \times 0.735818\ldots^2 = 0.4291$

44 c)

Curtis either hits the bullseye or does not hit the bullseye. The event that Curtis hits the bullseye is independent of every other time Curtis hits the bullseye. The probability of hitting the bullseye remains constant at 0.15.

45 a)

$\frac{2240}{3500} = 0.64$

$X \sim \mathrm{B}(24, 0.64)$

$\mathrm{P}(X = 15) = 0.1644$

45 b)

The probability of having a graphical calculator in a single class may not be constant at 64% as students may influence each other's choice to purchase one.

46 a)

$X \sim \mathrm{B}(30, 0.025)$

$\mathrm{P}(X = 0) = 0.4679$

46 b)

$\mathrm{P}(X \geq 3) = 1 - \mathrm{P}(X \leq 2) = 0.0384$

46 c)

A tile being faulty is independent of any other tile being faulty. The tiles are either faulty or they are not faulty. The probability of each tile being faulty is fixed at 0.025.

47 a)

$X \sim \mathrm{B}(40, 0.12)$

$\mathrm{P}(X = 4) = 0.1901$

47 b)

$\mathrm{P}(3 \leq X \leq 8) = \mathrm{P}(X \leq 8) - \mathrm{P}(X \leq 2)$

$= 0.955689\ldots - 0.126087\ldots = 0.8296$

47 c)

$\mathrm{P}(\text{Cinema club} \cap \text{Book club})$

$= \mathrm{P}(\text{Cinema club}) \times \mathrm{P}(\text{Book club})$

$= 0.3 \times 0.12 = 0.036$

47 d)

$X \sim \mathrm{B}(40, 0.036)$

$\mathrm{P}(X = 1) = 0.3446$

48)

$X \sim \mathrm{B}(15, \frac{3}{16})$

$\mathrm{P}(X = 2) = \binom{15}{2} \times (\frac{3}{16})^2 \times (\frac{13}{16})^{13}$

$= 105 \times \frac{13}{16} \times (\frac{3}{16})^2 \times (\frac{13}{16})^{12}$

$= \frac{1365}{16} \times (\frac{3}{16})^2 \times (\frac{13}{16})^{12}$

$= 455 \times \frac{3}{16} \times (\frac{3}{16})^2 \times (\frac{13}{16})^{12}$

$= 455 \times (\frac{3}{16})^3 \times (\frac{13}{16})^{12}$

$= \binom{15}{3} \times (\frac{3}{16})^3 \times (\frac{13}{16})^{12} = \mathrm{P}(X = 3)$

49 a)

$X \sim \mathrm{B}(7, 0.13)$

$\mathrm{P}(X = 3) = 0.0441$

49 b)

$Y \sim \mathrm{B}(7, 0.044053)$

$\mathrm{P}(Y \leq 1) = 0.9648$

50 a)

$X \sim \mathrm{B}(40, \frac{1}{8})$

$\mathrm{P}(X \geq 8 \mid X < 11) = \frac{\mathrm{P}(8 \leq X < 11)}{\mathrm{P}(X < 11)} = \frac{\mathrm{P}(X \leq 10) - \mathrm{P}(X \leq 7)}{\mathrm{P}(X \leq 10)}$

$= \frac{0.991655... - 0.880927...}{0.991655...} = 0.1117$

50 b)

$\mathrm{P}(X \geq 8 \mid X > 5) = \frac{\mathrm{P}(X \geq 8)}{\mathrm{P}(X > 5)} = \frac{1 - \mathrm{P}(X \leq 7)}{1 - \mathrm{P}(X \leq 5)} = \frac{1 - 0.880927...}{1 - 0.616221...}$

$= 0.3103$

50 c)

$\mathrm{P}(X \geq 8 \mid 5 < X < 11) = \frac{\mathrm{P}(8 \leq X < 11)}{\mathrm{P}(5 < X < 11)} = \frac{\mathrm{P}(X \leq 10) - \mathrm{P}(X \leq 7)}{\mathrm{P}(X \leq 10) - \mathrm{P}(X \leq 5)}$

$= \frac{0.991655... - 0.880927...}{0.991655... - 0.616221...} = 0.2949$

51 a)

Let X be the number of colour printers in use.

$X \sim \mathrm{B}(4, 0.15)$

$\mathrm{P}(X = 1) = 0.3685$

51 b)

Let Y be the number of black and white printers in use.

$Y \sim \mathrm{B}(8, 0.35)$

$\mathrm{P}(X \leq 2) = 0.4278$

51 c)

$\mathrm{P}(X = 0 \cap Y = 0) = \mathrm{P}(X = 0) \times \mathrm{P}(Y = 0)$

$= 0.522006\ldots \times 0.031864\ldots = 0.0166$

51 d)

$\mathrm{P}(X = 2 \cap Y = 0) + \mathrm{P}(X = 1 \cap Y = 1) +$

$\mathrm{P}(X = 0 \cap Y = 2)$

$= \mathrm{P}(X = 2) \times \mathrm{P}(Y = 0) + \mathrm{P}(X = 1) \times \mathrm{P}(Y = 1) +$

$\mathrm{P}(X = 0) \times \mathrm{P}(Y = 2)$

$= 0.097537\ldots \times 0.031864\ldots +$

$0.368475\ldots \times 0.137262\ldots +$

$0.522006\ldots \times 0.258686\ldots$

$= 0.00311 + 0.05058 + 0.13504 = 0.1887$

52)

$\mathrm{P}(X \leq 30) = 0.0912 < 0.1$

$\mathrm{P}(X \leq 31) = 0.1279 > 0.1$

So $k = 30$

53)

$\mathrm{P}(X < 65) = \mathrm{P}(X \leq 64) = 0.0923 < 0.1$

$\mathrm{P}(X < 66) = \mathrm{P}(X \leq 65) = 0.1157 > 0.1$

So $k = 65$

54)

$X \sim \mathrm{B}(300, 0.1)$

$\mathrm{P}(X < 19) = \mathrm{P}(X \leq 18) = 0.0097 < 0.01$

$\mathrm{P}(X < 20) = \mathrm{P}(X \leq 19) = 0.0171 > 0.01$

So $k = 19$

55)

$X \sim \mathrm{B}(400, 0.35)$

$\mathrm{P}(X > 155) = 1 - \mathrm{P}(X \leq 155) = 0.0529 > 0.05$

$\mathrm{P}(X > 156) = 1 - \mathrm{P}(X \leq 156) = 0.0427 < 0.05$

So $k = 156$

56)

$\binom{n}{0} \times 0.07^0 \times 0.93^n = 0.00465$

$\Rightarrow 0.93^n = 0.00465$

$\Rightarrow n = \log_{0.93} 0.00465 = 74.00905\ldots$

So $n = 74$

57)

$\binom{n}{0} \times 0.03^0 \times 0.97^n = 0.02921$

$\Rightarrow 0.97^n = 0.02921$

$\Rightarrow n = \log_{0.97} 0.02921 = 115.9992\ldots$

So $n = 116$

58)

$\binom{n}{n} \times 0.995^n \times 0.005^0 = 0.03983$

$\Rightarrow 0.995^n = 0.03983$

$\Rightarrow n = \log_{0.995} 0.03983 = 643.014\ldots$

So $n = 643$

59)

$\binom{n}{n} \times 0.974^n \times 0.026^0 = 0.01557$

$\Rightarrow 0.974^n = 0.01557$

$\Rightarrow n = \log_{0.974} 0.01557 = 158.002\ldots$

So $n = 158$

60)

$\binom{n}{0} \times 0.023^0 \times 0.977^n + \binom{n}{1} \times 0.023^1 \times 0.977^{n-1} = 0.39021$

$\Rightarrow 0.977^n + n \times 0.023 \times 0.977^{n-1} = 0.39021$

Use your calculator's SOLVE function

$\Rightarrow n = 89.0007\ldots$

So $n = 89$

61)

$\binom{n}{n-1} \times 0.998^{n-1} \times 0.002^1 + \binom{n}{n} \times 0.998^n \times 0.002^0 = 0.72913$

$\Rightarrow n \times 0.998^{n-1} \times 0.002 + 0.998^n = 0.72913$

Use your calculator's SOLVE function

$\Rightarrow n = 509.001\ldots$

So $n = 509$

62)

$X \sim \mathrm{B}(n, \frac{1}{15})$

$\mathrm{P}(X \geq 1) \geq 0.95 \Rightarrow 1 - \mathrm{P}(X = 0) \geq 0.95$

$\Rightarrow \mathrm{P}(X = 0) \leq 0.05 \Rightarrow \binom{n}{0} \times (\frac{1}{15})^0 \times (\frac{14}{15})^n \leq 0.05$

$\Rightarrow (\frac{14}{15})^n \leq 0.05 \Rightarrow \ln(\frac{14}{15})^n \leq \ln 0.05$

$\Rightarrow n \ln(\frac{14}{15}) \leq \ln 0.05 \Rightarrow n \geq \frac{\ln 0.05}{\ln(\frac{14}{15})} = 43.42\ldots$

So 44 donors must be selected for there to be at least a 95% chance of at least one having the blood type B positive.

63)

$X \sim \mathrm{B}(n, \frac{1}{70})$

$\mathrm{P}(X \geq 1) \geq 0.99 \Rightarrow 1 - \mathrm{P}(X = 0) \geq 0.99$

$\Rightarrow \mathrm{P}(X = 0) \leq 0.01 \Rightarrow \binom{n}{0} \times (\frac{1}{70})^0 \times (\frac{69}{70})^n \leq 0.01$

$\Rightarrow (\frac{69}{70})^n \leq 0.01 \Rightarrow \ln(\frac{69}{70})^n \leq \ln 0.01$

$\Rightarrow n \ln(\frac{69}{70}) \leq \ln 0.01 \Rightarrow n \geq \frac{\ln 0.01}{\ln(\frac{69}{70})} = 320.05\ldots$

So 321 components must be selected for there to be at least a 99% chance of at least one being faulty.